

## Multiple Choice (10 marks)

1. What is the dimensionality of the null space of the following matrix?  $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ 
  - (a) 0
  - (b) 1
  - (c) 2
  - (d) 3
2. Statement 1| The original ResNets and Transformers are feedforward neural networks. Statement 2| The original Transformers use self-attention, but the original ResNet does not.
  - (a) True, True
  - (b) False, False
  - (c) True, False
  - (d) False, True
3. Neural networks:
  - (a) Optimize a convex objective function
  - (b) Can only be trained with stochastic gradient descent
  - (c) Can use a mix of different activation functions
  - (d) None of the above
4. What is the rank of the following matrix?  $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ 
  - (a) 0
  - (b) 1
  - (c) 2
  - (d) 3
5. Statement 1| RELUs are not monotonic, but sigmoids are monotonic. Statement 2| Neural networks trained with gradient descent with high probability converge to the global optimum.
  - (a) True, True
  - (b) False, False
  - (c) True, False
  - (d) False, True

## True / False (10 marks)

*Answer True or False*

6. True or False: As the number of training examples increases, the variance of the model's predictions tends to decrease.
7. True or False: Both Bayesians and frequentists agree on the use of prior distributions for parameters in all types of probabilistic models.
8. True or False: In Naive Bayes, attributes are statistically independent of one another given the class value.
9. True or False: In the given Bayesian network, 15 independent parameters are required if no assumptions about independence are made.
10. True or False: High entropy indicates that the partitions in classification are not pure, leading to greater uncertainty in predictions.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to check the given decimal with a precision of 2 by using regex.
12. Write a python function to count the number of rectangles in a circle of radius  $r$ .

*End of Paper*